DSA Experiment 6

Title: Implement stack using linked list

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```
#include  <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node* next;
};
struct Node* top=NULL;
void push(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    if (newNode==NULL) {
        printf("Stack Overflow!\n");
        return;
    }
    newNode->data=value;
    newNode->next=top;
    top=newNode;
    printf("Pushed %d\n", value);
}
void pop() {
    if (top==NULL) {
        printf("Stack Underflow! The stack is empty.\n");
        return;
    }
    struct Node* temp=top;
    printf("Popped %d\n",top->data);
    top=top->next;
    free(temp);
}
void display() {
    if (top == NULL) {
        printf("Stack is empty.\n");
        return;
    }

    struct Node* temp = top;
    printf("Stack elements: ");
```

```
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}
int main() {
    push(10);
    push(77);
    push(100);
    display();
    pop();
    display();
    return 0;
}
```

OUTPUT:

Output

```
Pushed 20
Pushed 68
Pushed 200
Stack elements: 200 68 20 NULL
Popped 200
Stack elements: 68 20 NULL
```